

9.4.18 in large (90/20)

Question: Consider a random sample $x_1, \dots, x_n$ from a two-parameter Weibull density

$$f(x) = \frac{K}{\lambda} \left(\frac{x}{\lambda}\right)^{K-1} e^{-(x/\lambda)^K}$$

over the interval $(0, \infty)$. Show that the maximum of the loglikelihood with respect to $\lambda > 0$ for $K > 0$ fixed satisfies $\hat{\lambda}^K = \frac{1}{n} \sum_{i=1}^n x_i^K$.

Derive the profile loglikelihood $\ln L(K, \lambda) = \ln \left(\prod_{i=1}^n \frac{K}{\lambda} \left(\frac{x_i}{\lambda}\right)^{K-1} e^{-(x_i/\lambda)^K} \right) + (K-2) \sum_{i=1}^n \ln x_i + \text{constant}$

by substituting this λ into the ordinary loglikelihood. Since the second term of the profile loglikelihood is constant in K , one can apply the regression (9.3).

Prove that this leads to the MM-estimator

$$\frac{1}{K+2} = \frac{\sum_{i=1}^n x_i^K \ln x_i}{\sum_{i=1}^n x_i^K} = \frac{1}{n} \sum_{i=1}^n \ln x_i$$

First off, to get the profile loglikelihood, we take the Weibull distribution, and take the logarithm of it to see where that gets us.

$$f(x) = \frac{K}{\lambda} \left(\frac{x}{\lambda}\right)^{K-1} e^{-(x/\lambda)^K}$$

now, our likelihood function is product of individual densities.

$$L(K, \lambda) = f(x_1) \times \dots \times f(x_n) = \prod_{i=1}^n f(x_i, K, \lambda) = \prod_{i=1}^n \frac{K}{\lambda} \left(\frac{x_i}{\lambda}\right)^{K-1} e^{-(x_i/\lambda)^K}$$

So we go over every x_n in our random sample.

next step, take the logarithm

$$\begin{aligned} \ln L(K, \lambda) &= \ln \left(\prod_{i=1}^n \left(\frac{K}{\lambda} \left(\frac{x_i}{\lambda}\right)^{K-1} e^{-(x_i/\lambda)^K} \right) \right) \\ &= \sum_{i=1}^n \ln \left(\frac{K}{\lambda} \left(\frac{x_i}{\lambda}\right)^{K-1} e^{-(x_i/\lambda)^K} \right) \\ &= \sum_{i=1}^n \left[\ln \left(\frac{K}{\lambda} \right) + \ln \left(\left(\frac{x_i}{\lambda}\right)^{K-1} \right) + \ln \left(e^{-(x_i/\lambda)^K} \right) \right] \\ &= \sum_{i=1}^n \left[\ln(K) - \ln(\lambda) + (K-1) \left(\ln(x_i) - \ln(\lambda) \right) - (x_i/\lambda)^K \right] \\ &= \sum_{i=1}^n \left[\ln(K) - \ln(\lambda) + (K-1) \ln(x_i) - K \ln(\lambda) - (x_i/\lambda)^K \right] \\ &= \sum_{i=1}^n \left[\ln(K) + (K-2) \ln(x_i) - K \ln(\lambda) - (x_i/\lambda)^K \right] \end{aligned}$$

From here, I sum over each element as we have constants K and λ .

$$\begin{aligned} \ln L(K, \lambda) &= \sum_{i=1}^n \ln(K) + \sum_{i=1}^n (K-2) \ln(x_i) - \sum_{i=1}^n K \ln(\lambda) - \sum_{i=1}^n (x_i/\lambda)^K \\ &= n \ln(K) + (K-2) \sum_{i=1}^n \ln(x_i) - nK \ln(\lambda) - \frac{1}{\lambda^K} \sum_{i=1}^n x_i^K \end{aligned}$$

This gives the loglikelihood as

$$\ell(K, \lambda) = n \ln(K) - nK \ln(\lambda) + (K-2) \sum_{i=1}^n \ln(x_i) - \frac{1}{\lambda^K} \sum_{i=1}^n x_i^K$$

Now I must show the max loglikelihood with respect to $\lambda > 0$ for $K > 0$ fixed satisfies $\hat{\lambda}^K = \frac{1}{n} \sum_{i=1}^n x_i^K$

So this is the previous step we done with respect to λ

$$\frac{\partial \ell}{\partial \lambda} = \frac{\partial}{\partial \lambda} \left[n \ln(K) - nK \ln(\lambda) + (K-2) \sum_{i=1}^n \ln(x_i) - \frac{1}{\lambda^K} \sum_{i=1}^n x_i^K \right] \Rightarrow \frac{\partial}{\partial \lambda} \left[-nK \ln(\lambda) - \frac{1}{\lambda^K} \sum_{i=1}^n x_i^K \right]$$

Setting the terms:

$$\frac{\partial \ell}{\partial \lambda} = 0 - \frac{nK}{\lambda} + 0 + \frac{K}{\lambda^{K+1}} \sum_{i=1}^n x_i^K$$

$$\frac{\partial \ell}{\partial \lambda} = -\frac{nK}{\lambda} + \frac{K}{\lambda^{K+1}} \sum_{i=1}^n x_i^K$$

To get max $\frac{\partial \ell}{\partial \lambda} = 0$

$$0 = -\frac{nK}{\lambda} + \frac{K}{\lambda^{K+1}} \sum_{i=1}^n x_i^K$$

We manipulate equation

$$\frac{nK}{\lambda} = \frac{K}{\lambda^{K+1}} \sum_{i=1}^n x_i^K \Rightarrow \frac{n}{\lambda} = \frac{1}{\lambda^{K+1}} \sum_{i=1}^n x_i^K \Rightarrow n \lambda^K = \sum_{i=1}^n x_i^K \Rightarrow \lambda^K = \frac{1}{n} \sum_{i=1}^n x_i^K$$

This gives what we needed

$$\hat{\lambda}^K = \frac{1}{n} \sum_{i=1}^n x_i^K$$

Now the profile log likelihood.

We want to get $\ln L(K, \hat{\lambda}) = \ln L(K, \frac{1}{n} \sum_{i=1}^n x_i^K) + (K-2) \sum_{i=1}^n \ln x_i + \text{constant}$

So we substitute $\hat{\lambda} = \left(\frac{1}{n} \sum_{i=1}^n x_i^K \right)^{1/K}$

$$\ell(K, \hat{\lambda}) = n \ln(K) - nK \ln(\hat{\lambda}) + (K-2) \sum_{i=1}^n \ln(x_i) - \frac{1}{\hat{\lambda}^K} \sum_{i=1}^n x_i^K$$

Let's take together

$$\begin{aligned} \ell(K) &= n \ln(K) - n \ln \left(\left(\frac{1}{n} \sum_{i=1}^n x_i^K \right)^{1/K} \right) + n \ln(n) + (K-2) \sum_{i=1}^n \ln(x_i) - n \\ &\text{we group } n \ln(n) - n \text{ as constant. } x_i^K = e^{K \ln x_i} \end{aligned}$$

$$\ell(K) = n \ln(K) - n \ln \left(\frac{1}{n} \sum_{i=1}^n e^{K \ln x_i} \right) + (K-2) \sum_{i=1}^n \ln(x_i) + \text{constant}$$

log rules:

- 1 $\ln(ab) = \ln(a) + \ln(b)$
- 2 $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$
- 3 $\ln(a^b) = b \ln(a)$
- 4 $\ln(e^a) = a$
- 5 $\ln\left(\prod a_i\right) = \sum \ln(a_i)$ ← b this a rule?

$$a) \frac{\partial}{\partial \lambda} n \ln(K) = 0 \text{ as constant!}$$

$$b) \frac{\partial}{\partial \lambda} -nK \ln(\lambda) = -nK \frac{\partial}{\partial \lambda} \ln(\lambda) = -\frac{nK}{\lambda}$$

$$c) \frac{\partial}{\partial \lambda} (K-2) \sum_{i=1}^n \ln(x_i) = 0 \text{ as also a constant!}$$

$$d) \frac{\partial}{\partial \lambda} -\frac{1}{\lambda^K} \sum_{i=1}^n x_i^K = -\left(\sum_{i=1}^n x_i^K \right) \frac{\partial}{\partial \lambda} \frac{1}{\lambda^K} = -\left(\sum_{i=1}^n x_i^K \right) \frac{\partial}{\partial \lambda} \lambda^{-K} = -\left(\sum_{i=1}^n x_i^K \right) (-K) \lambda^{-K-1} = \frac{K}{\lambda^{K+1}} \sum_{i=1}^n x_i^K$$

$$a) n \ln(K) \text{ unchanged}$$

$$\begin{aligned} b) -nK \ln(\hat{\lambda}) &= -nK \ln \left(\left(\frac{1}{n} \sum_{i=1}^n x_i^K \right)^{1/K} \right) \text{ log rule!} \\ &= -nK \frac{1}{K} \ln \left(\frac{1}{n} \sum_{i=1}^n x_i^K \right) = -n \ln \left(\frac{1}{n} \sum_{i=1}^n x_i^K \right) \\ &= -n \left[\ln \left(\sum_{i=1}^n x_i^K \right) - \ln(n) \right] \\ &= -n \ln \left(\sum_{i=1}^n x_i^K \right) + n \ln(n) \end{aligned}$$

$$c) (K-2) \sum_{i=1}^n \ln(x_i) \text{ unchanged}$$

$$d) -\frac{1}{\hat{\lambda}^K} \sum_{i=1}^n x_i^K = -\frac{n}{\sum_{i=1}^n x_i^K} \cdot \sum_{i=1}^n x_i^K = -n$$

How have the prior log likelihood:

$$\ell(K) \ln(K) - \ln \ln \left(\sum_{i=1}^m e^{K \ln x_i} \right) + (K-1) \sum_{i=1}^m \ln x_i + \text{constant}$$

So as per hint, diff this with respect to K

$$\begin{aligned} \frac{d\ell}{dK} &= \frac{d}{dK} [\ln(K)] - \frac{d}{dK} \left[\ln \left(\sum_{i=1}^m e^{K \ln x_i} \right) \right] + \frac{d}{dK} \left[(K-1) \sum_{i=1}^m \ln x_i \right] + \frac{d}{dK} [\text{constant}] \\ &= \frac{1}{K} - \frac{\sum_{i=1}^m x_i^K \ln x_i}{\sum_{i=1}^m x_i^K} + \sum_{i=1}^m \ln x_i \end{aligned}$$

Set this to zero for score equation $K = G_m$

$$0 = \frac{1}{K} - \frac{\sum_{i=1}^m x_i^K \ln x_i}{\sum_{i=1}^m x_i^K} + \sum_{i=1}^m \ln x_i = 0$$

$$\frac{1}{K} = \frac{\sum_{i=1}^m x_i^K \ln x_i}{\sum_{i=1}^m x_i^K} - \sum_{i=1}^m \ln x_i$$

$$\frac{1}{K} = \frac{\sum_{i=1}^m x_i^K \ln x_i}{\sum_{i=1}^m x_i^K} - \frac{1}{m} \sum_{i=1}^m \ln x_i$$

Hint: Set $K_{max} = G(K_m)$

$$G(K) = \frac{1}{K} = \frac{\sum_{i=1}^m x_i^K \ln x_i}{\sum_{i=1}^m x_i^K} - \frac{1}{m} \sum_{i=1}^m \ln x_i$$

So Kupper updates can be got

$$\frac{1}{K_{max}} = \frac{\sum_{i=1}^m x_i^{K_{max}} \ln x_i}{\sum_{i=1}^m x_i^{K_{max}}} - \frac{1}{m} \sum_{i=1}^m \ln x_i$$

It's maybe not elegant, but I at least attempted to work through it.

Question 3:

Show that if $U \sim \text{Uniform}(0,1)$

$$X = \lambda [-\log(1-U)]^{1/K} \sim \text{Weibull}(K, \lambda)$$

$$\text{CDF} = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a, b] \\ 1 & \text{for } x > b \end{cases}$$

$$\text{PDF} = \begin{cases} \frac{K}{\lambda} \left(\frac{x}{\lambda} \right)^{K-1} e^{-(x/\lambda)^K} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\text{CDF} = F(x) = \begin{cases} 1 - e^{-(x/\lambda)^K} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

So I see that its quantile (inverse of CDF) is

$$Q(p) = \lambda (-\ln(1-p))^{1/K}$$

this later solves.

Im going to start with the CDF and reduce/simplify equation.

$$\begin{aligned} F(x) &= 1 - e^{-(x/\lambda)^K} \\ \downarrow \\ 1 - F(x) &= e^{-(x/\lambda)^K} \\ \downarrow \\ \ln(1 - F(x)) &= -(x/\lambda)^K \\ \downarrow \\ \ln(1 - u) &= -(x/\lambda)^K \\ \downarrow \\ -\ln(1 - u) &= (x/\lambda)^K \\ \downarrow \\ [-\ln(1 - u)]^{1/K} &= x/\lambda \\ \downarrow \\ x &= \lambda [-\ln(1 - u)]^{1/K} \end{aligned}$$

So Weibull distribution from Uniform?

$$\frac{d}{dK} \left[\ln \ln \left(\sum_{i=1}^m e^{K \ln x_i} \right) \right]$$

We have chain rule

first order 2 param.

$$S_1 \frac{d\theta}{dK} = \dot{\theta}(g(K)) \cdot g'(K)$$

$$\text{outer } f(u) = \ln(u)$$

$$\text{inner } u = \sum_{i=1}^m x_i^K$$

$$S_1 \frac{d}{dK} \ln(u) \cdot \frac{d}{dK} \left(\sum_{i=1}^m x_i^K \right)$$

$$\frac{1}{u} = \frac{1}{\sum_{i=1}^m x_i^K} \quad \frac{d}{dK} \left(\sum_{i=1}^m x_i^K \right) = \sum_{i=1}^m \frac{d}{dK} (x_i^K) = \sum_{i=1}^m x_i^K \ln(x_i)$$

together giving what was shown in the question

$$M \frac{\sum_{i=1}^m x_i^K \ln x_i}{\sum_{i=1}^m x_i^K}$$